

Session 1: RDBMS Fundamentals, DDL & DML Commands

Professional Solutions

Question: 1. Create COVID Database Schema

Create Database

```
CREATE DATABASE covid_db;  
USE covid_db;
```

Create Tables

```
CREATE TABLE covid_cases (  
country VARCHAR(50),  
report_date DATE,  
confirmed_cases INT,  
deaths INT,  
recoveries INT,  
population BIGINT  
);
```

```
CREATE TABLE covid_deaths (  
country VARCHAR(50),  
report_date DATE,  
total_deaths INT,  
death_rate DECIMAL(5,2)  
);
```

```
CREATE TABLE covid_vaccines (  
country VARCHAR(50),  
report_date DATE,  
vaccinated_people BIGINT,  
total_doses BIGINT  
);
```

Question: 2. Insert Initial COVID Data

```
INSERT INTO covid_cases  
(country, report_date, confirmed_cases, deaths, recoveries, population)  
VALUES  
( 'India', '2023-01-01', 44680000, 530000, 44100000, 1400000000),  
( 'India', '2023-02-01', 44720000, 530500, 44150000, 1400000000),  
( 'USA', '2023-01-01', 103000000, 1120000, 100000000, 331000000),  
( 'USA', '2023-02-01', 103500000, 1125000, 100500000, 331000000),  
( 'UK', '2023-01-01', 24500000, 220000, 24200000, 67000000),  
( 'UK', '2023-02-01', 24600000, 221000, 24300000, 67000000);
```

Question: 3. Update Case Information

```
UPDATE covid_cases
SET confirmed_cases=44750000
WHERE country='India'
AND report_date='2023-02-01';
```

Question: 4. Delete Incorrect Records

```
DELETE FROM covid_cases
WHERE country='IN';
```

```
DELETE FROM covid_cases
WHERE country='India'
AND report_date='2023-01-01'
LIMIT 1;
```

Question: 5. Alter Table to Add Vaccination Information

```
ALTER TABLE covid_cases
ADD vaccination_rate DECIMAL(5,2);
```

```
UPDATE covid_cases SET vaccination_rate=76.50 WHERE country='India';
```

```
UPDATE covid_cases SET vaccination_rate=81.20 WHERE country='USA';
```

```
UPDATE covid_cases SET vaccination_rate=79.40 WHERE country='UK';
```